

# **Community Behaviour & Participation Analysis on Welfare Outreach**

## **NSS Open Projects 2026**

### **Track 5 – Analytics & Insights**

#### **Challenge 5.2 – Community Behaviour & Participation Analysis on Welfare Schemes**

### **Problem Statement**

Public welfare schemes in India often face uneven participation across districts. Differences in literacy, education levels, workforce participation, and demographic conditions can influence awareness and accessibility of welfare initiatives.

This project aims to identify districts that may require targeted welfare outreach by analysing district-level Census 2011 indicators and constructing a Welfare Readiness Score.

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### **Dataset**

Source: Census 2011 District-Level Dataset

Coverage:

- 640 districts
- 25 demographic and socio-economic indicators

Key variables:

- Population
- Literacy
- Workforce participation

- Educational attainment
  - Gender distribution
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## **Methodology**

### **Data Cleaning**

The dataset was checked for missing values and inconsistencies.

### **Feature Engineering**

The following indicators were derived:

- Literacy Rate
- Worker Participation Rate
- Female Ratio
- Graduate Education Rate

### **Welfare Readiness Score**

A composite score was developed using:

- Literacy Rate (40%)
- Worker Rate (30%)
- Graduate Rate (30%)

Min-Max scaling was applied before score computation.

### **District Segmentation**

Districts were classified into:

- High Readiness
- Moderate Readiness
- Low Readiness
- Very Low Readiness

using quartile-based segmentation.

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## Key Findings

1. Significant variation exists across districts.
  2. Literacy and graduate education show strong positive relationships with welfare readiness.
  3. Several districts exhibit consistently low readiness scores and may require targeted intervention.
  4. Educational attainment appears to be an important factor affecting welfare awareness and access.
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## Intervention Recommendations

For districts with the lowest readiness scores:

- Literacy and awareness campaigns
  - Community outreach programs
  - Welfare enrollment assistance camps
  - Educational support initiatives
  - Targeted information dissemination through local institutions
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## Limitations

This analysis uses demographic indicators as proxies for welfare participation because district-level welfare scheme enrollment data was not available.

Future work may integrate PM-KISAN, Ayushman Bharat, PMUY, or MGNREGA datasets to directly evaluate welfare uptake.

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## Conclusion

The proposed framework identifies districts that may require additional welfare outreach and support. The methodology is reproducible and can be extended using scheme-specific participation datasets in future studies.